

CircuitLab

Agents that find which attention heads implement a behavior in GPT-2 small, counting every forward pass.

1. Challenge tackled

Mapping the circuit behind a model behavior usually means patching every component by hand. We asked whether a team of agents can do this search under a hard compute budget, and checked them against a known answer: the 26-head IOI circuit (arXiv:2211.00593). The users are interpretability researchers who want a cheap, auditable first pass over a new behavior.

2. Tools / ML models used

- **Claude Sonnet** - Planner and analyst. **Claude Haiku** - literature scout, task designer, hypothesis, experimentalist and safety agents.
- **Omnigent** - agent orchestration, handoffs and policies (claim gate, sweep gate, spend cap).
- **MCP tool server** - one shared server, so all agents use one forward-pass counter.
- **TransformerLens 3.9.0 + PyTorch** - GPT-2 small, activation patching, mean ablation, path patching, faithfulness.
- **Streamlit** - live view of the research record. **arXiv search** - literature priors.

3. What worked well

Method	Forward passes	Result
Exhaustive patching, 144 heads	292	8 published circuit heads in the top 10
Agent run 1	366	L9H9, L10H0, L9H6; faithfulness 0.8827 vs 0.0275 random; recall 3/26; human approved the claim
ACDC (arXiv:2304.14997)	not recorded	cited, not re-run (needs TransformerLens 1.6.1, torch<2)

- The guardrails fired as designed: a human approved the run 1 claim, approved sweeps over 58 and 101 heads and refused one over 31 heads in run 3, and the spend cap stopped the models at \$1.50 in run 1.
- A joint-only path-patching mode (about 8 passes per sender instead of about 28) and a 24-sender limit make an upstream trace affordable.
- In run 3 the agents proposed 14 early-layer senders, 4 of them real upstream circuit heads (L0H10, L2H2, L4H11, L5H8), against about 1.5 expected by chance. These were not confirmed by path patching.

4. What was challenging

- **Cost:** an open-ended upstream sweep (60 senders x 10 receivers) cost about 1700 passes, six times the exhaustive baseline. We stopped it and added the joint-only mode and the 24-sender limit.
- **Stopping rule:** faithfulness saturates with 3 heads, so it is a poor reason to stop. Run 3 required an upstream trace, but it ended before presenting a claim.
- **Fairness:** the analyst could call `score_vs_truth`, which leaks counts from the answer key, and the hypothesis agent recalled the blocked IOI paper from memory. Recall in runs 1 and 3 is optimistic. We removed the tool for future runs.
- **Setup:** ACDC's pinned environment conflicts with ours; Windows needed `Git sh` on `PATH` and UTF-8 mode for Omnigent.
- Honest bottom line: the agents have not beaten the exhaustive baseline on passes. One seed (42) only.

5. How we spent our time

- 2026-10-03: chose circuit discovery on GPT-2 small and wrote the build plan.
- 2026-10-04, from 12:23 to 15:45 (logged): IOI dataset, patching, ablation, path patching, faithfulness, benchmarks and the exhaustive baseline.
- 2026-10-04, in parallel: Omnigent setup, agent specs, claim and sweep gates, tool server, Streamlit view, three agent runs.
- 2026-10-04, evening: run report, results write-up and submission. Hours per person: not recorded.

If we had 24 more hours

We would enforce hard per-experiment pass limits in code, run a 5-seed recovery curve (recall against passes for agents, exhaustive and random), try a behavior with no published circuit, and run real ACDC in a pinned environment.